

14-8.26

classmate

Date

Page

DAY - 12

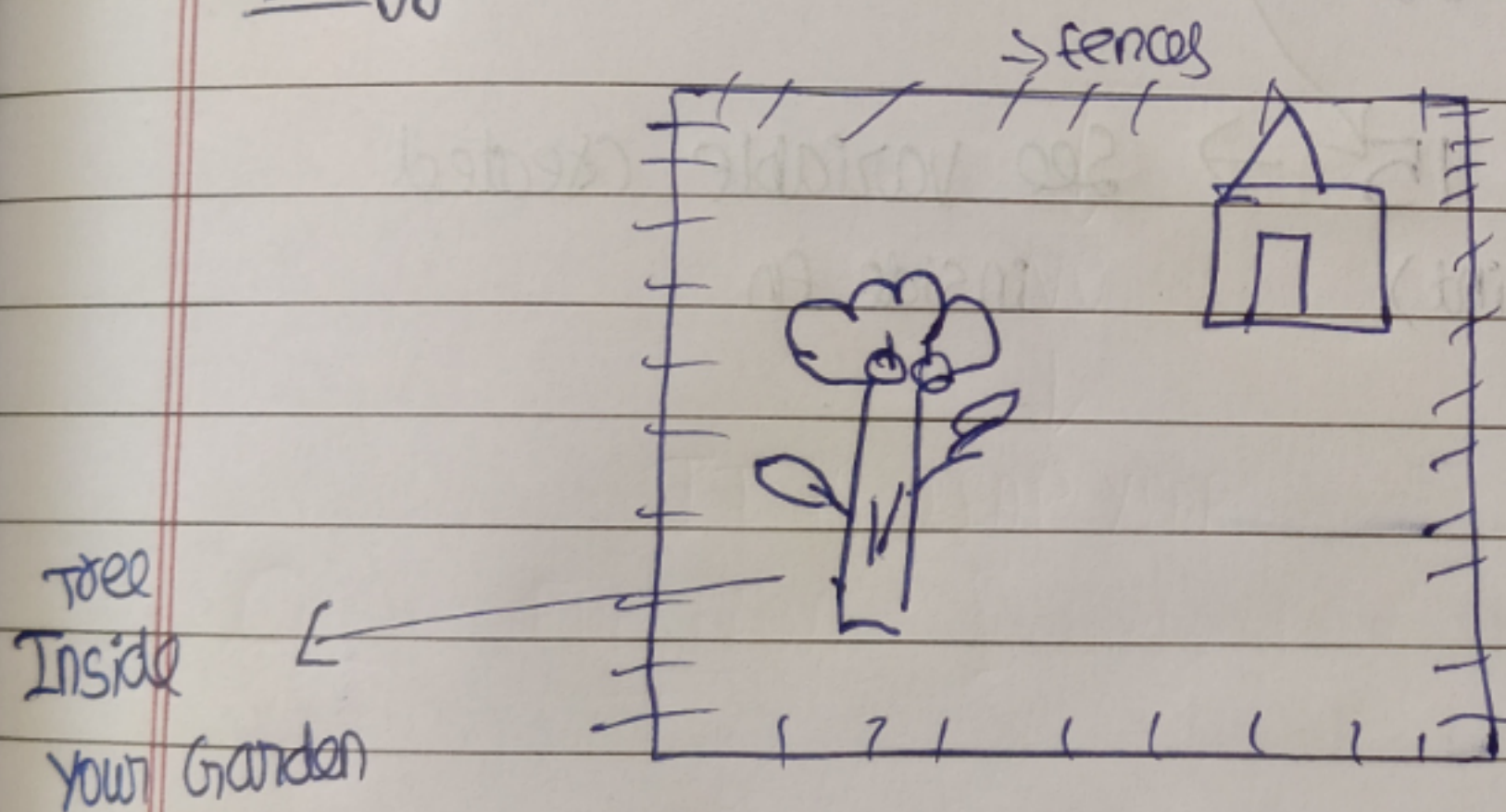
SCOPE

I. Name - Spaces : Local vs Global Scope :

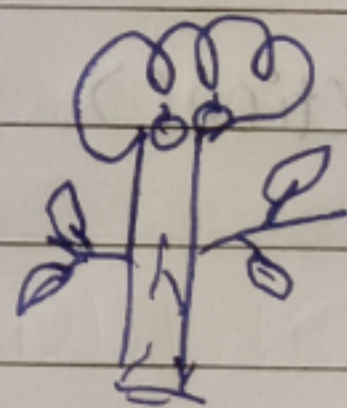
What is mean by scope??

It means, area of a program where a variable or name is accessible.

Analogy :



so, only u can
access apples in it



↳ Tree Outside Your
Garden. Any
outsiders can take
Apple from it.

Types of Scope in Python

* Python follows the LEGB Rule :

1. Local - scope (L)

It means [a variable created inside a function] ^(X)
Its scope stays local.

Example :

```
def my_function():
```

```
    Point = 15 → See variable created inside fn  
    print(Point)
```

```
my_function()
```

OUTPUT :

15

But ; what if i print outside ?

```
def my_function():  
    Point = 15
```

```
my_function()
```

```
print(Point)
```

#Error

OUTPUT : NameError: name Point is not defined

NOTE - 1 :
indeed

that variable
we're

* Because,

2. Enclosing

It is
and inner

" when

The o
inner f
fun var

Ex :

```
def outerc()
```

```
    outerc()
```

OUTPUT :

15

Date _____
Page _____

NOTE - 1 : When you create a new variable or indeed a new function inside another function, that variable is only accessible / u can use, only when we're inside function.

* Because, it has a local-scope

* for nested fn case (otherwise it is local scope concept only)

2. Enclosing Scope (E) :

Enclosing Scope Variable mean :

It is a variable which is created inside the outer function and inner function can access that variable.

" When a function is inside another function,

The outer function ^(scope) variable is the enclosing scope for inner function : which means inner function can access outer fun variable it can change.

Ex:

```
def Outer():
```

```
    x = 15    # x → belongs to outer scope
```

```
    def inner():
```

```
        print(x)
```

→ nested function
→ due to it is nested inside fn variable we can use "Local Scope/variable"

```
    inner()
```

Specifically, Here it called Enclosing Scope

```
Outer()
```

x belongs to Outer(), but inner() can access it.

OUTPUT:

∴ x is enclosing scope of inner().

15

Q. What if we wanted that inside accessible outside the function?

↓ leads to

3. Global Scope (G)

Global scope means ; It's a variable created outside all functions.

* It is written at top level part of Py file.

Example :

Point = 15 — Global Scope

```
def score():  
    print(Point)
```

score()

OUTPUT :

15

Important Rule :

* If u assign to a variable inside a function, Python normally treats that variable as local.

Point = 15

```
def change():
```

Point = 20

change()

print(Point)

→ If i put print here then here also it prints

why?

inside ch

* If you d

use

Example :

point = 1

def cha

Change C

print C

OUTPUT

20

why?

Point = 20 created a new local variable inside change(). It didn't change global point.

* If you actually want to modify the global variable, use global → keyword (X)

Example: Special CASE:

→ Point = 15 (but) (X)

```
def change():  
    global Point  
    Point = 20
```

global Change variable Point = 15 to 20 it says
global Point
Variable name

Change()

Print(Point)

OUTPUT:

20

Here: Global scope :

Ex-1:

Player-health = 10

def drink_Potion():

Potion_strength = 2

print(Player-health)

drink_Potion()

OUTPUT:

10

Ex-2: Player-health = 10

def drink_Potion():

global Player-health

Player-health = 15

print(Player-health)

drink_Potion()

print(Player-health)

OUTPUT:

15

15

Both are same here bcoz, only one variable

II. DOES PYTHON HAVE [BLOCK SCOPE] ?

Block-scope mean : Block is a section of code controlled by S.T lik

if

for

while

- * In some languages like C, C++, Java, a variable created inside that block exists only inside block.

FOR EX:

if cond :

x = 10 $\longleftrightarrow$ x belongs only to this block

outside : x doesn't even exist

" It is called Block-scope "

- * Python is Different :

if (True) :

x = 10

Print(x)

Output : 10

" Here ; x is still accessible outside if-block "

But functions DO create scope:

```
def test():  
    x = 10  
test()  
print(x)
```

* Here 'x' is inaccessible because, function creates local scope.

∴ PYTHON HAS FUNCTION SCOPE, NOT ORDINARY BLOCK-SCOPE.